



Python for Data Science - 2305CS303

Lab - 4

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1) Character Category Counter

Problem: Count uppercase, lowercase, digits, and special characters in a string.

Sample Inputs & Outputs:

- Input: Python3@Lab → Uppercase: 2, Lowercase: 7, Digits: 1, Special: 1
- Input: HELLO123 → Uppercase: 5, Lowercase: 0, Digits: 3, Special: 0
- Input: hi@#9 → Uppercase: 0, Lowercase: 2, Digits: 1, Special: 2

```
In [6]: i="Python3@Lab"
i2="HELLO123"
i3="HI@#9"
u=0
l=0
d=0
s=0
for ch in i3:
    if ch.isupper():
        u+=1
    elif ch.islower():
        l+=1
    elif ch.isdigit():
        d+=1
    else:
        s+=1
print("count uppercase: ",u," count lowercase: ",l," digits: ",d," special: ",s)

count uppercase:  2  count lowercase:  0  digits:  1  special:  2
```

2) Reverse Each Word

- Input: Python is Easy → nohtyP si ysaE

- Input: Hello World → olleH dlroW
- Input: Learn Python Fast → nraeL nohtyP tsaF

```
In [25]: n=input("enter string: ").split()
for ch in range(len(n)):
    n[ch]=n[ch][-1::-1]
n=" ".join(n)
n
```

Out[25]: 'nohtyp si ysae'

3) Case Pattern Identifier

- PYTHON → Uppercase
- python → Lowercase
- Python Programming → Title Case

```
In [27]: n=input("enter str: ")
if n.isupper():
    print("Uppercase")
elif n.islower():
    print("lowercase")
elif n.istitle():
    print("title case")
```

Uppercase

4) First and Last Occurrence Finder

- programming , g → First:3 Last:10
- banana , a → First:1 Last:5
- hello , z → Not found

```
In [32]: n=input("enter str: ")
n1=input("enter char: ")
f=n.find(n1)
l=n.rfind(n1)
if f== -1 and l== -1:
    print("Not found")
else:
    print(f,l)
```

Not found

5) Word Frequency Counter (ignore the case)

- Python is easy and Python is powerful → python: 2
- Java is popular but java is verbose → java: 2
- C is fast → c:1

```
In [34]: n=input("enter str: ").lower()
n1=input("enter word: ").lower()
count=n.count(n1)
print(count)
```

2

6) Remove Extra Spaces

- Python is fun → Python is fun
- Hello World → Hello World
- NoExtraSpace → NoExtraSpace

```
In [40]: n=input("enter str: ")
n=" ".join(n.split())
n
```

```
Out[40]: 'Python is fun'
```

7) Prefix and Suffix Removal

- unhappy.txt → happy
- pretest.py → test
- unwanted.doc → wanted

```
In [43]: n=input("enter str: ")
pre=input("enter prefix: ")
suf=input("enter suffix: ")
n=n.removeprefix(pre)
n=n.removesuffix(suf)
print(n)
```

```
happy
```

8) Replace Vowels

- Programming → Pr*gr*mm*ng
- Education → *d*c*t**n
- Sky → Sky

```
In [60]: n=input("enter str: ")
n1="aeiou"
n2=""
for ch in n:
    if ch in n1:
        n2+="*"
    else:
        n2+=ch
n2
```

```
Out[60]: 'Pr*gr*mm*ng'
```

9) String Compression

- aaabbccccc → a3b2c4
- xxxyyyz → x3y3z1
- abcd → a1b1c1d1

```
In [6]: n=input("enter str: ")
count=0
l=""
for i in range(1,len(n)):
    if n[i-1]==n[i]:
        count+=1
    else:
        l+=n[i-1]+str(count)
        count=1
if n:
    l += n[-1] + str(count)
l
```

Out[6]: 'a2b3c3'

10) Toggle Case (without using str.swapcase())

- PyTh0n → pYtHoN
- HELlo → heLLo
- Python → pYTHON

```
In [63]: n=input("enter str: ")
n2=""
for ch in n:
    if ch.isupper():
        n2+=ch.lower()
    elif ch.islower():
        n2+=ch.upper()
n2
```

Out[63]: 'pYTHON'

11) Username Validation

valid username rules:

1. only alphabets and digits
2. does not start with a digit
3. length ≥ 6

- User123 → Valid
- 123User → Invalid
- Us@12 → Invalid

```
In [81]: n=input("enter str: ")
if n.isalnum():
    if n[0].isdigit():
        print("Invalid")
    else:
        if len(n)>=6:
            print("Valid")
        else:
            print("Invalid")
```

```
else:
    print("Invalid")
```

Valid

12) Palindrome Checker (ignore the case)

- Madam → Palindrome
- Level → Palindrome
- Python → Not Palindrome

```
In [87]: n=input("enter string: ")
n1=n[::-1]
if n==n1:
    print("Palindrome")
else:
    print("not Palindrome")
```

Palindrome

13) Longest Word Finder

- Python programming is interesting → programming
- I love coding → coding
- Data structures and algorithms → structures

```
In [101... n=input("enter string: ").split()
max=[]
max2=""
for ch in range(len(n)):
    if len(n[ch])>=len(max2):
        max=ch
        max2=n[ch]
n[max]
```

Out[101... 'algorithms'

14) Case Conversion

- python programming
 - Upper: PYTHON PROGRAMMING
 - Lower: python programming
 - Title: Python Programming
 - Capitalize: Python programming

```
In [102... n=input("enter str: ")
print(n.upper())
print(n.lower())
print(n.title())
print(n.capitalize())
```

PYTHON PROGRAMMING
python programming
Python Programming
Python programming

15) Custom Split

- apple,banana,grape
- 10|20|30
- A-B-C-D

In [107...

```
n=input("enter str: ")  
n1=input("enter split char: ")
```